

AYMANE AIT DADS

Software Dev Engineer Intern - Machine Learning | ML Pipeline Design · LoRA Fine-Tuning · Scalable ML Systems

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PROFESSIONAL SUMMARY

Ranked **17th out of 3,000+ teams** in the NVIDIA Nemotron Reasoning Challenge (score 0.86/1.0, \$106K+ prize pool) by designing and owning an end-to-end LoRA SFT pipeline on a 30B-parameter model - directly demonstrating the scalable ML engineering expected of this Amazon Robotics role. At Orange Maroc, built production-grade Random Forest and K-Means models on 100K+ network records, reducing manual analysis time by 60% and delivering outputs consumed directly by the network optimization team. Brings hands-on experience in PyTorch, BF16 mixed-precision training, ablation study design, and full ML pipeline ownership - aligning precisely with Amazon Robotics' need for engineers who build reliable, customer-impacting systems at scale. Aims to apply rigorous model evaluation and scalable ML engineering practices to Amazon Robotics' real-world automation challenges, contributing to systems that directly impact customer experience.

CORE COMPETENCIES

Machine Learning Engineering | Scalable ML Pipelines | Model Evaluation & Ablation | Deep Learning (PyTorch) | Data-Driven Optimization | Feature Engineering | End-to-End Pipeline Ownership | Collaborative Team Environment

WORK EXPERIENCE

Orange Maroc

Jul 2024 - Aug 2024

Data Science Intern - Casablanca, Morocco

- Built a **Random Forest + K-Means pipeline** on 100K+ fiber-optic network records (upload speed, latency, jitter) to classify network quality across thousands of nodes, delivering outputs directly consumed by the network optimization team.
- Developed an **automated clustering pipeline** mapping customer performance profiles to 5 commercial service tiers, eliminating manual categorization and scaling analysis across the full node inventory.
- Reduced manual analysis time by **~60%** through automated feature engineering and model evaluation workflows built in Python, Pandas, and Scikit-learn.
- Delivered **stakeholder presentations** translating model outputs into actionable maintenance recommendations, bridging ML results to business decision-making for non-technical audiences.

PROJECTS

NVIDIA Nemotron Model Reasoning Challenge

Kaggle Competition · In Progress (June 2026) · Ranked 17th / 3,000+ Teams

- Ranked **17th / 3,000+ teams** (score 0.86/1.0, \$106K+ prize pool) in the NVIDIA Nemotron Reasoning Challenge by fine-tuning a **30B-parameter Mamba-Transformer** with LoRA SFT (r=32, BF16, completion-only loss, 8,192-token context, grad_accum=32) across 5 structured reasoning categories.
- Ran **controlled ablation experiments** across LR schedules, packing strategies, and synthetic augmentation - discovering that reasoning-style consistency outweighs data volume in LLM SFT; built custom cipher parsers and 8-bit operation solvers to produce 605 verified cipher rows and 64 verified bit-manipulation rows for clean CoT training data.

PyTorch · Unsloth · TRL · PEFT · Hugging Face Transformers · vLLM · Triton · Kaggle GPU

AI-Generated Image Detection

EURECOM ImSecu + NTIRE 2026 @ CVPR · 2025 · Ranked 1st in Class

- Ranked **1st in the EURECOM class leaderboard** (0.791 AUC) by fine-tuning a CLIP ViT-L/14 backbone (428M params) with a custom MLP head, dual learning rates (2e-6 / 5e-4), and a 3-model ensemble with 10-view TTA on 250K images spanning 25+ generator types.
- Submitted independently to the **NTIRE 2026 @ CVPR CodaBench** international benchmark; established through ablation that 1-epoch training (0.793 AUC) outperforms 4-epoch training (0.767 AUC), demonstrating superior out-of-distribution generalization.

PyTorch · OpenCLIP · ViT · FP16 Mixed Precision · AdamW · Kaggle

EDUCATION

Engineering Degree in Data Science (Master-level) - EURECOM

Sep 2023 - Present

Relevant coursework: Machine Learning, Deep Learning, Computer Vision, NLP, Statistics, Cloud Computing, Image Security, Web Applications

CPGE - Mathematics & Physics - Ibn Timiya

2021 - 2023

SKILLS

ML Frameworks & Engineering: PyTorch | Scikit-learn | Hugging Face Transformers | Unsloth | TRL | PEFT | vLLM

LLM & Model Training: LoRA | SFTTrainer | BF16/FP16 Mixed Precision | Completion-only Loss | Ablation Studies | Chain-of-Thought | OpenCLIP

Data Science & Infrastructure: Python | Pandas | NumPy | Feature Engineering | SQL | Git | Docker (basic) | Power BI